

AUTOMATED CLASS SCHEDULING WEBSITE

Project Report

Submitted By

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Abstract

The Automated Class Scheduling Website is a web-based application developed to simplify and automate the process of timetable generation for educational institutions. Traditional timetable creation requires significant manual effort and often results in scheduling conflicts, inefficient classroom utilization, and faculty overlaps.

This application allows administrators to manage courses, faculty members, classrooms, and automatically generate optimized schedules based on predefined constraints. The generated timetable can also be viewed, filtered, and exported for further use.

The project demonstrates the practical implementation of modern frontend technologies while solving a real-world scheduling problem.

Introduction

Educational institutions often spend many hours preparing class timetables manually. Managing multiple classrooms, faculty members, and courses while avoiding scheduling conflicts becomes increasingly difficult as the institution grows.

The Automated Class Scheduling Website addresses these challenges by providing an intelligent scheduling platform capable of generating optimized class schedules efficiently.

The system reduces manual work, minimizes scheduling errors, improves resource utilization, and enhances overall academic planning.

Objectives

- Automate class timetable generation.
- Reduce scheduling conflicts.
- Manage faculty availability efficiently.
- Allocate classrooms according to capacity and resources.
- Improve classroom utilization.
- Save administrative time.
- Provide downloadable timetable reports.
- Ensure an easy-to-use interface.

Problem Statement

Manual timetable preparation suffers from several problems:

- Faculty timing conflicts
- Classroom allocation conflicts
- Uneven workload distribution
- Human errors
- Time-consuming scheduling process
- Difficulty updating schedules

The proposed system automates these tasks using scheduling constraints and validation techniques.

Scope of the Project

The system is suitable for:

- Colleges
- Universities
- Training Institutes
- Schools
- Coaching Centers

Future enhancements may include:

- Student login
- Faculty login
- Database integration
- AI-based timetable optimization
- Attendance management
- Cloud deployment

Technologies Used

Frontend

- React.js
- TypeScript
- Vite
- Tailwind CSS

UI Components

- shadcn/ui
- Radix UI
- Lucide React Icons

Development Tools

- Visual Studio Code
- npm
- Git
- GitHub

The project is configured as a Vite React application using React, TypeScript, and modern UI component libraries.

System Modules

1. Course Management

The Course Management module allows administrators to:

- Add new courses

- Assign course codes
- Set academic year (FY, SY, TY)
- Allocate faculty
- Specify prerequisites
- Define classroom requirements
- Set enrollment limits
- Manage course duration

The module maintains complete course information required for schedule generation.

2. Faculty Management

This module manages faculty information including:

- Faculty name
- Email
- Department
- Specialization
- Weekly workload
- Available teaching hours

Faculty availability is stored for each weekday to avoid scheduling conflicts.

3. Classroom Management

The classroom module stores:

- Classroom name
- Building
- Seating capacity
- Room type
- Available resources
- Weekly availability

Resources such as projectors, laboratories, smart boards, Wi-Fi, and audio systems can be assigned to classrooms.

4. Schedule Generator

The Schedule Generator is the core component of the application.

It validates scheduling constraints such as:

- Faculty availability
- Classroom capacity
- Required classroom resources
- Course prerequisites
- Teacher assignment
- Scheduling conflicts

The generator also provides:

- Progress indicator
- Constraint validation

- Efficiency statistics
- Conflict detection

5. Schedule Viewer

After generation, schedules can be:

- Viewed in Grid View
- Viewed in List View
- Filtered by Year
- Filtered by Faculty
- Filtered by Classroom
- Exported as CSV

This makes timetable distribution simple and organized.

Features

- User-friendly interface
- Responsive design
- Course management
- Faculty management
- Classroom management
- Automatic schedule generation
- Conflict detection
- Constraint validation
- Statistics dashboard
- Schedule filtering
- CSV export
- Modern UI components

Scheduling Constraints

The scheduling algorithm checks:

- Faculty availability
- Classroom availability
- Room capacity
- Required resources
- Course prerequisites
- Assigned faculty
- Time slot availability

If any constraint fails, the system displays warnings or errors before generating the timetable.

Advantages

- Saves time

- Reduces human errors
- Avoids timetable conflicts
- Efficient classroom utilization
- Better faculty workload management
- Easy timetable updates
- Professional reporting
- Scalable architecture

Limitations

- Uses sample data
- No authentication system
- No backend database
- No cloud synchronization
- No mobile application

Future Enhancements

- Login authentication
- MySQL/PostgreSQL integration
- AI timetable optimization
- Attendance management
- Student dashboard
- Faculty dashboard
- Notifications
- PDF timetable export
- Mobile application
- Cloud deployment

System Screenshots

The following screenshots illustrate the application's user interface, including the main dashboard, the schedule generation results with conflict detection, and the grid-based schedule viewer.

1. Dashboard and Course Management

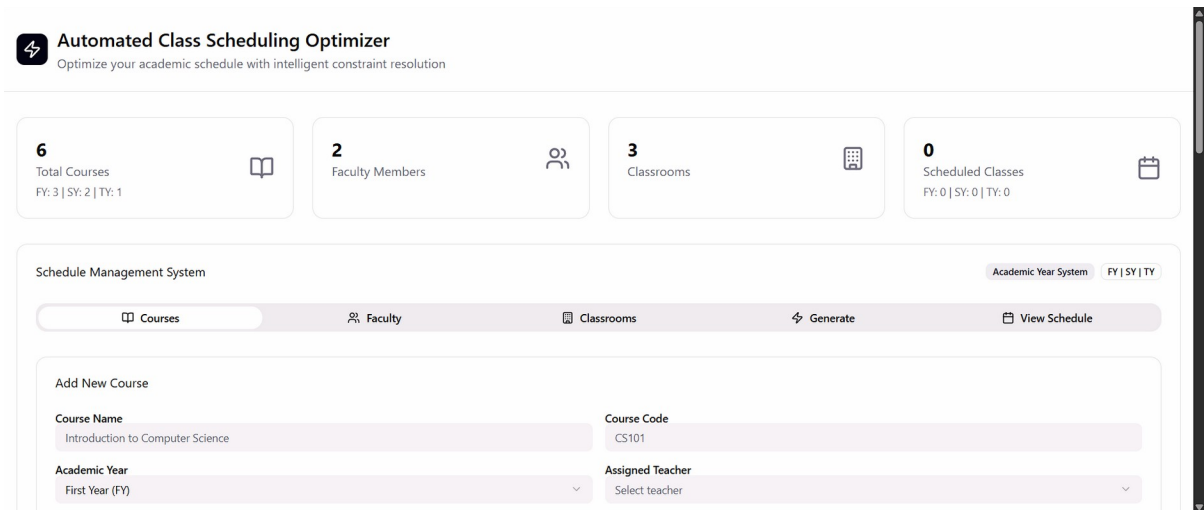


Figure 1: Main dashboard showing course, faculty, and classroom summary along with the Add New Course form.

2. Generated Schedule with Conflict Detection

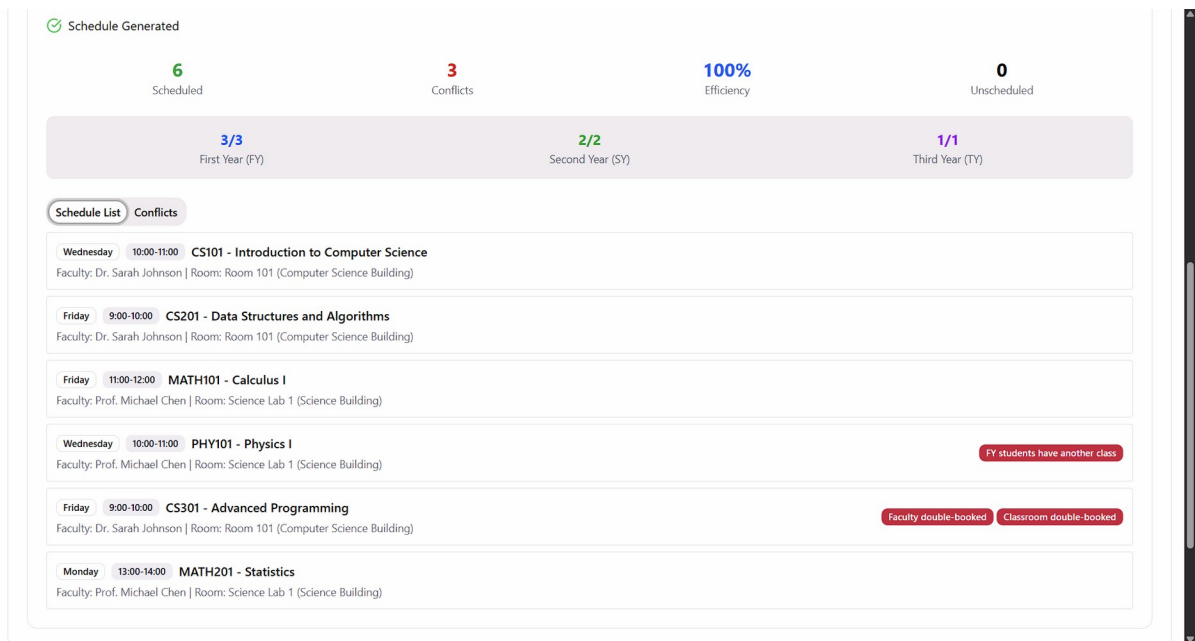


Figure 2: Schedule generation results showing scheduling statistics, efficiency percentage, and detected conflicts such as faculty and classroom double-booking.

3. Grid View of Weekly Schedule

Grid View

List View

By Year

First Year (FY)

Time	Monday	Tuesday	Wednesday	Thursday	Friday
8:00-9:00					
9:00-10:00					
10:00-11:00			CS101 FY Dr. Sarah Johnson Room 101 PHY101 FY Prof. Michael Chen Science Lab 1 Conflict		
11:00-12:00					MATH101 FY Prof. Michael Chen Science Lab 1
12:00-13:00					

Figure 3: Grid View of the First Year (FY) weekly schedule, with a highlighted scheduling conflict.

Conclusion

The Automated Class Scheduling Website successfully demonstrates how modern web technologies can simplify one of the most time-consuming administrative tasks in educational institutions.

By automating timetable generation, validating scheduling constraints, managing institutional resources, and providing export functionality, the system offers a reliable and scalable solution for academic scheduling. The project also highlights the practical application of React, TypeScript, and modern UI frameworks in developing real-world management systems.